

DRAFT · PRE-MAINNET

REINEIRA LABS LIMITED

ReineiraOS Lightpaper

Confidential settlement, RSS, and the ReineiraOS builder stack in one read

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Abstract

Encrypted-state conditional settlement on programmable chains lacks a primitive surface that composes pluggable verification with pluggable underwriting under a shared operator set. ReineiraOS specifies four composable primitives (confidential escrow, pluggable gate, FHE-priced recourse, slashing-secured operator set) over a TFHE coprocessor substrate, formalises them as the Reineira Settlement Standard (RSS), and ships them with an AI-native builder stack — Reineira Atlas and Reineira Code. The protocol launches under a Public-Infrastructure regime: every contract is an immutable singleton with no upgrade authority; the protocol charges no fees; there is no token; settlement is non-custodial; operators stake cUSDC and earn their own relay fees. The Reineira Settlement Protocol is published by Reineira Labs Limited (RAK DAO Free Zone) as a Software Vendor. Arbitrum L2 is the canonical deployment; Ethereum L1 is planned; OP-Stack and Atlas-deployed third-party L3s extend the host inventory. Chaos-net deploys on Arbitrum L2 in July 2026.

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1. Abstract

Encrypted-state conditional settlement on programmable chains lacks a primitive surface that combines pluggable verification and underwriting under a shared Operator set. ReineiraOS specifies four composable primitives over a TFHE coprocessor substrate: confidential Escrow, pluggable Gate, FHE-priced Recourse, and a slashing-secured Operator set. It formalises these primitives as the Reineira Settlement Standard (RSS) and ships them with an AI-native builder stack — Reineira Atlas and Reineira Code. The protocol launches under a **Public-Infrastructure regime**: every contract is an immutable singleton with no upgrade authority (§11.1); the protocol charges no fees; there is no token; settlement is non-custodial; and Operators stake cUSDC and earn their own relay fees (§8.8). Reineira Labs Limited (RAK DAO Free Zone) publishes the Reineira Settlement Protocol as a Software Vendor (§11.2). Arbitrum L2 is the canonical deployment, Ethereum L1 is planned, and OP-Stack and Atlas-deployed third-party L3s extend the host inventory (§9). Chaos-net deploys on Arbitrum L2 in July 2026.

2. Problem & Scope

Two prior research programmes have matured in parallel without converging. Encrypted-state EVMs (Aztec, Penumbra, Zama fhEVM, Inco, Fhenix CoFHE) make confidential computation tractable at contract granularity. Conditional-release patterns (HTLCs, OpenZeppelin escrows, Sablier streams, parametric coverage at Nexus and Sherlock) separate the release predicate from the value held. No prior system composes encrypted conditional release, pluggable verification, pluggable underwriting, and a single Operator set under cross-graph slashing as one primitive surface. That gap is the protocol's subject (see whitepaper §1.1).

The protocol's load-bearing contributions, as developed across §§5–13 of the whitepaper, are:

1. **The Reineira Settlement Standard (RSS).** A conformance specification covering interface coverage, encrypted-state typing, silent-failure indistinguishability, namespaced upgrade-safe storage, and canonical events. It is separate from the Reference Implementation and uses semver-versioned releases and a Reineira Improvement Proposal process (see whitepaper §5).
2. **A four-primitive composable surface.** Confidential Escrow, pluggable `IConditionResolver` Gates, FHE-priced Recourse with four-role separation, and a permissionless Operator set under stake-weighted task authorization. The primitives interoperate through `@reineira-os/shared` interfaces without modifying core contracts (see whitepaper §§6, 7, 8).
3. **The silent-failure pattern.** A control-flow construction that makes the success and failure transcripts of an encrypted-condition redemption computationally indistinguishable to any party without decryption authority. Proposition 4.3 formalises the pattern under TFHE IND-CPA (see whitepaper §4.5).
4. **A layer-asymmetric license vector with a pre-announced trigger schedule.** The core protocol converts from BUSL-1.1 to Apache 2.0 on the irrevocable Change Date 2029-06-01. Five publication-time accelerators may shorten, but cannot extend, the conversion path (see whitepaper §13).

The whitepaper specifies the protocol layer: the on-chain Settlement Protocol and the RSS conformance surface. The **builder stack** comprises Reineira Atlas (the venture scaffolding engine; twelve AI agents; /**bootstrap** scaffolds a venture from a one-page brief) and Reineira Code (AI-assisted resolver and policy authoring; /**new-resolver**, /**new-policy**, /**audit**). It is described where it supports the open-economy thesis but is not a normative surface. The internals of off-chain Operator services and Reineira Labs

Limited commercial products are out of scope. The protocol does not compete on fee minimisation, pure permissionlessness at the Recourse-pool layer, agentic micropayments at the under-ten-dollar tier, or L2 substrate performance — the substrate is consumed, not built (see whitepaper §1.4).

3. Architecture in one diagram

The protocol decomposes into five independently replaceable layers, each addressing one concern. Higher layers depend only on lower-layer interfaces; lower layers are unaware of which higher-layer caller invokes them (see whitepaper §3.2).

Layer	Concern	Representative contracts
Ap- pli- ca- tion	Public API: <code>create</code> , <code>fund</code> , <code>redeem</code> , <code>purchase</code> , <code>Coverage</code> , <code>dispute</code> , <code>invoke</code>	<code>ConfidentialEscrow</code> , <code>ConfidentialCoverageManager</code> , <code>AgentInvocationAdapter</code> , <code>@reineira-os/sdk</code>
FHE	Encrypted primitives — <code>euint64</code> , <code>eaddress</code> , <code>ebool</code> ; homomorphic arithmetic, comparison, conditional selection	Fhenix CoFHE precompiles, substrate input adapters
Storage	Encrypted state, safe upgrades via proxy, gasless user paths	ERC-7201 namespaced layouts, UUPS proxies, the protocol base contract (ERC-2771)
Or- ches- tra- tion	Task distribution, cross-chain relay, operator-fee routing, dispute arbitration	<code>OperatorRegistry</code> , <code>CoordinatorRegistry</code> , <code>TaskExecutor</code> , <code>FeeManager</code> , <code>OperatorSlashingManager</code>
Set- tle- ment	Cross-chain native stablecoin flow	<code>CCTPV2EscrowReceiver</code> (Circle CCTP V2 for USDC), <code>LZ0FTEscrowReceiver</code> (LayerZero OFT / USDT0 for USDT), <code>ConfidentialUSDC</code> , <code>ConfidentialUSDT</code>

The on-chain protocol is a pnpm monorepo with five Solidity-bearing packages and three supporting packages: `@reineira-os/shared` (base contracts, interfaces, mocks), `@reineira-os/tokens` (confidential wrappers), `@reineira-os/escrow`, `@reineira-os/recourse`, and `@reineira-os/orchestration`, plus off-chain services, plugin examples, and the SDK. The `shared` package is the only permitted cross-package Solidity import path; contracts cross package boundaries only through `@reineira-os/shared`. This rule keeps the storage and upgrade invariants enforceable across the monorepo (see whitepaper §3.3).

The protocol follows a *standards-fork pattern*. The standard layer (RSS, §5) is published free and OSI-aligned; the Reference Implementation (the contracts under `packages/`) ships under BUSL-1.1 with a Change Date of 2029-06-01 and Apache 2.0 as the Change License. Other parties may implement the standard in their own contract code. Conformance depends on satisfying the RSS interface and conformance tests, not on deploying the Reference Implementation (see whitepaper §3.9).

Above the protocol perimeter, ReineiraOS publishes a **builder stack** — Reineira Atlas (`reineira-atlas`) and Reineira Code (`reineira-code`) — that mechanises venture scaffolding and RSS plugin authoring. Atlas is a Claude Code repository with twelve AI agents organised across strategy, product, growth, execution, intelligence, and legal. From a one-page brief, `/bootstrap` generates a populated docs tree and a working application skeleton (`platform-modules/backend` + `platform-modules/app` with ZeroDev smart accounts). Reineira Code is the AI-assisted Solidity authoring environment: slash commands (`/new-resolver`, `/new-policy`, `/audit`, `/integrate`) generate RSS-conformant Solidity, run the cofhejs FHE test suite, and wire the result into the Atlas-generated application. Both repositories are platform-locked to a contract-interface version (`platform: "0.1"` in `reineira.json`), so the builder stack moves in lockstep with RSS (see whitepaper §1.1 for the layered diagram).

4. The four primitives

The protocol composes all functionality from four primitives. Whitepaper §4 gives the detailed cryptographic treatment of encrypted mode; §5 specifies the RSS layer that every conforming implementation must satisfy.

Escrow. The conditional-release primitive holds value, or the right to value, until a pluggable resolver confirms that the release condition has been met. The Reference Implementation, **ConfidentialEscrow**, stores encrypted **owner**, **caller**, **amount**, **paidAmount**, and **isRedeemed** state for each record and routes the redemption decision through the silent-failure pattern. Key invariant: encrypted amount conservation. For every funded Escrow, the contract’s encrypted token balance moves exactly with the per-Escrow **paidAmount** accumulators under additive homomorphism (see whitepaper §§6.1, 6.2, Proposition 6.2).

Gate / Condition Resolver. A condition resolver is an external contract implementing the pluggable release-Gate interface, **IConditionResolver** (with **isConditionMet(escrowId) → bool** and **onConditionSet(escrowId, data)**), plus ERC-165. An Escrow binds at most one resolver per **escrowId** at creation and consults it on **redeem** before evaluating the encrypted ownership predicate. A null resolver is the always-true condition. Key invariant: a true resolver result is necessary but not sufficient — the silent-failure path still hides any failure of the ownership or full-funding predicates (see whitepaper §§5.6, 6.5).

Recourse. This FHE-priced parametric coverage primitive lets independent underwriters deploy **IConfidentialUnderwriterPolicy** contracts and compete for LP capital. Coverage binds to an existing Escrow position, and the policy returns an encrypted risk score for homomorphic premium pricing. Key contracts: **ConfidentialCoverageManager**, **ConfidentialRecoursePool**, **ConfidentialPolicyRegistry**, **ConfidentialPoolFactory**. Distinct authorisation modifiers (**onlyUnderwriter**, **onlyCoverageManager**, **onlyOwner**) enforce four contract-layer roles: Policy Builder, Pool Creator, Pool Manager, and Operator. Key invariant: LP solvency. **payClaim** caps every payout in ciphertext against the pool’s encrypted total liquidity before debiting, so no claim can exceed funded capital (see whitepaper §§7.1, 7.4, Proposition 7.1).

Operators. Operators form the permissionless off-chain execution layer. They register in **OperatorRegistry**, post cUSDC stake, and claim and execute tasks dispatched by **TaskExecutor**. **FeeManager** pays Operators, while **OperatorSlashingManager** applies slashing. Operators perform work the host EVM cannot evaluate natively: cross-chain attestations (Circle CCTP V2 Iris, LayerZero DVN sets), FHE ciphertext settlement against the CoFHE coprocessor, and off-chain quorum verdicts that move asynchronous resolvers from **Pending** to **Resolved**. Key invariant: slashing safety. Every stake decrement traces to an on-chain **SlashProposal** whose final state is **Executed** and whose **evidence** hash is recorded in the corresponding event (see whitepaper §§8.1, 8.2, Proposition 8.1).

5. RSS — standard vs reference implementation

The Reineira Settlement Standard (RSS) defines conformance for confidential settlement across the four primitives. The contracts under **packages/** are one *Reference Implementation*; other parties may publish conforming implementations using their own contract code (see whitepaper §5.1).

Publishing the standard separately from the implementation is a load-bearing commitment. A competing settlement contract can implement RSS, legitimising the standard and inheriting its plugin ecosystem, or fork it, fragmenting the addressable plugin and Operator surface relative to the conforming majority. The protocol charges nothing; the open standard and the Reference Implementation are complementary rather than substitutable, and each requires the other to remain coherent (see whitepaper §5.1).

RSS covers the four-primitive interface surface and canonical interfaces; encrypted-type contract conventions and the silent-failure pattern; upgrade and storage conventions (ERC-7201 namespacing, __

gap[50] reserves, UUPS proxies, ERC-2771 meta-transactions); and the cross-chain transport binding through which any bridge funnels into the Escrow funding sink. RSS excludes fee policy, execution-stack choices for third-party self-hosted deployments, and the builder stack used to author conforming plugins (see whitepaper §5.2). RSS is semver-versioned: **v0.1** ships at testnet (June 2026), followed by **v0.2** at chaos-net (Q4 2026) with the first five Reineira Improvement Proposals (see whitepaper §5.3).

A deployment is RSS-conformant if and only if it satisfies clauses (a)–(f) below (see whitepaper §5.4):

- (a) **Interface coverage.** The deployment implements every core and plugin interface declared mandatory for its mode (public or encrypted).
- (b) **Encrypted state.** In encrypted mode, the deployment stores every owner identifier, payment amount, paid-amount accumulator, and condition-derived intermediate as an encrypted type per Definition 4.1. In public mode, it runs the same contracts with plaintext types under the dual-mode design, preserving the interface shape so plugins and integrations remain binary-compatible across modes.
- (c) **Silent-failure pattern.** Every redemption path that evaluates an encrypted condition conforms to Proposition 4.3: its success and failure transcripts (events, gas, storage diffs, ciphertext payloads) are computationally indistinguishable to a PPT adversary without decryption authority over the relevant intermediates.
- (d) **Namespaced upgrade-safe storage.** All upgradeable contracts use ERC-7201 namespaced storage per Definition 3.1, inherit from the protocol base contract, carry a `uint256[50] private __gap`, and follow the EIP-1822 UUPS proxy pattern over the EIP-1967 proxy-slot layout.
- (e) **Meta-transaction support.** Sponsored user paths route through an ERC-2771 trusted forwarder, allowing gasless flows without compromising authorisation by recovering `msg.sender` from the appended forwarder context rather than the originating EOA.
- (f) **Canonical events.** The deployment emits the events declared in `ICore` and `IEscrowEvents` (and the analogous canonical event sets for the Recourse and Operator surfaces), allowing cross-implementation indexers, slashing oracles, and bridge-receipt monitors to observe state transitions without depending on implementation-specific event schemas.

6. Confidentiality model

Permission model. Ciphertexts are addressable on-chain by handle, but the substrate allows only addresses on a ciphertext’s access list to recover its plaintext. Contract calls create and revoke access-list entries (`FHE.allow`, `FHE.allowTransient`, `FHE.allowThis`). The coprocessor enforces this metadata at decryption time; EVM code does not. Selective disclosure to an auditor, counterparty, or regulator applies the same primitive to a specific recipient. Granting `allow` to a designated recipient role lets that role — and no other observer — request decryption without changing the on-chain ciphertext or any other observer’s view. The set of EVM addresses authorised to request decryption of a ciphertext is its *Decryption Authority* (Definition 4.2; see whitepaper §4.4).

Silent-failure pattern. This is the protocol’s load-bearing confidentiality design. When a state-mutating call would otherwise leak its outcome through a revert, the contract evaluates the decision predicate homomorphically and combines the per-condition `ebool` handles into one encrypted Boolean, `eValid`. It then uses `eValid` to select homomorphically between the intended effect and a no-op, producing identical on-chain traces for success and failure. Reverting on a decryption-revealed condition would publish its evaluation to every chain observer and defeat the confidentiality goal. Proposition 4.3 (Silent-Failure Indistinguishability) formalises the pattern: for any probabilistic polynomial-time adversary without decryption authority, the advantage in distinguishing success and failure transcripts is negligible in the substrate security parameter, conditional on TFHE IND-CPA (see whitepaper §4.5).

Observable vs confidential. Some state must remain plaintext: event topics consumed by indexers, gas-payer addresses billed by the EVM, Operator identities tied to slashing collateral, and sequential Escrow IDs used by downstream cross-chain handlers. This split is intentional and encoded in the contract surface (see whitepaper §4.6). The quantities deliberately exposed by the protocol form its *Public Skeleton* (Definition 4.4).

Quantity	Observable (plaintext)	Confidential (ciphertext)
Escrow existence	escrow ID (sequential)	owner, caller, amount, paid amount, redemption
Funding	funder address (when not intermediated)	funded amount, fee, premium
Redemption	event emission, sender	recipient, amount, success / failure outcome
Coverage	pool address, policy address	holder, coverage amount, risk score

7. Multi-chain hosting

Canonical host — Arbitrum L2. Chaos-net deploys to Arbitrum L2 in mid-July 2026, followed by v1.0 mainnet hardening across Q4 2026 (with the first independent Operator-set audit complete by the end of Q4 2026). Arbitrum L2 remains the canonical host across all roadmap horizons. Three properties support this choice: Fhenix CoFHE — the FHE coprocessor on which encrypted mode depends — targets Arbitrum-compatible deployment; Arbitrum is a mature L2 with a settled fee market and deep stablecoin liquidity; and the standard Arbitrum L1↔L2 gateway lists L1-issued ERC-20s (including the confidential stablecoin wrappers) on L2 without bespoke governance ceremony (see whitepaper §9.7).

Planned and long-term hosts. Ethereum L1 is planned as a second host for institutional and L1-finality use cases and remains public-mode-only until an L1 FHE coprocessor lands. OP-Stack hosting is on the long-term roadmap; activation timing and vendor remain TBD pending OP ecosystem maturity, and OP-Stack native bridging is tracked as Open Problem 9.1. The protocol does not migrate away from Arbitrum L2; additional hosts run in parallel (see whitepaper §§9.8, 9.9).

Atlas-deployed third-party L3s. Third-party rollups — including Arbitrum Orbit L3s settling to Arbitrum One and, when supported, OP-Stack L3s — extend ReineiraOS coverage by **self-deploying** the protocol through Reineira Atlas. The L3 owner uses Atlas’s **/bootstrap** and protocol-deployment skills to scaffold an RSS-conformant deployment, then maintains it on its own cadence. Reineira Labs Limited neither operates nor upgrades these deployments. The L3 requires **liquidity** from the ReineiraOS network: Operators registered on Arbitrum L2 route cross-chain funding into L3-deployed Escrows over the parent chain’s **native rollup bridge** (Arbitrum L2→L3 bridge for Orbit; OP-Stack native bridge TBD). Self-deployed L3s do not ship their own CCTP V2 or LayerZero OFT handlers because those transports operate at the L1/L2 layer; the native rollup bridge is the only required transport (see whitepaper §9.10).

8. Public-Infrastructure launch posture

The protocol launches under a Public-Infrastructure regime. Five structural commitments hold from chaos-net Day 1:

1. **Immutable contracts.** Every protocol contract is an immutable singleton deployed at a fixed address, with no upgrade authority or admin keys. Functional evolution uses parallel deployments at new addresses; users opt in to migration (see whitepaper §11.1).
2. **The protocol charges nothing.** Before and after mainnet, there is no protocol fee, commission, or treasury. **FeeManager** is an **operator-fee router**, not a protocol-revenue contract: it computes an executing Operator’s relay fee and pays that Operator. No activation block can switch on a protocol

- charge, and no one can toggle a protocol-fee slot. Gate builders, Recourse pools and their liquidity providers, and Operators set and keep their own fees; Reineira receives none (see whitepaper §8.8).
3. **cUSDC operator bond.** Operators stake cUSDC, an immutable ERC-7984 wrapper of USDC. The Operator economy is functional and slashable without requiring any token (see whitepaper §8.1).
 4. **Software Vendor, non-custodial.** Reineira Labs Limited (RAK DAO Free Zone, UAE) publishes the source-available software as Software Vendor. It does not operate the protocol, custody user funds, or hold admin keys (see whitepaper §11.2).
 5. **Permissionless from Day 1.** Operator admission is stake-and-sanctions-only at the contract layer. Any sanctions-clean address meeting the cUSDC-denominated `minStake` threshold can register, regardless of jurisdiction. There is no curated allowlist or admission gate to sunset (see whitepaper §8.4).

There is **no token** in the v1.0 design. No protocol-issued instrument underwrites any surface, and the Operator economy remains cUSDC-bonded throughout. The protocol's economics are fully specified without one (see whitepaper §12).

9. Licensing

The core protocol source is published under Business Source License 1.1 with Change Date ~~2029-06-01~~ and Apache License 2.0 as the Change License. **Licensors at publication: Reineira Labs Limited** (RAK DAO Free Zone), the Software Vendor that publishes the source-available software. The Change Date is irrevocable and announced when the license is published. On that date, the entire BUSL-licensed corpus converts to Apache License 2.0 through the BUSL Change License clause, without further action by the Licensors. This advance commitment allows the trigger schedule to move conversion earlier, never later (see whitepaper §13.1).

Different protocol layers carry different licenses at publication. Builder-facing layers (interfaces, SDK, token wrappers, plugin examples) are permissively licensed to reduce licensing friction for integrators. Defensive layers (Escrow, Recourse, orchestration core) carry BUSL-1.1 during the contestability window and convert to Apache License 2.0 on the Change Date. The full layer-asymmetric license vector — Apache 2.0 for **shared** and off-chain Operator services, MIT for token wrappers and the SDK, CC-BY-4.0 for documentation, BUSL-1.1 → Apache 2.0 for the core — is given in Tab. 13.1 (see whitepaper §13.2).

The Licensors commit to five triggers announced at license publication. Each accelerates conversion to Apache License 2.0 but cannot delay it. Advance publication prevents later license tightening by removing the Licensors' discretion to relicense more restrictively (see whitepaper §13.4).

10. Risk & audit posture

This lightpaper and the companion whitepaper are pre-mainnet drafts. Both stabilise as edition **v1.0** only at chaos-net mainnet (Q4 2026), in lockstep with contract v1.0 and the first external smart-contract audit. Chaos-net ships without a completed external audit. Published security evidence comprises an internal Phase 7 self-review with ~85 manual findings and eight remediated mainnet blockers, Slither / Aderyn tool baselines, and a Safe-plus-Timelock deployment posture (whitepaper §10.6). Users assume the risk of using chaos-net deployments during the June 2026 → Q4 2026 window. External-audit completion is committed for the Q4 2026 v1.0 hardening pass; subsequent editions will revise this section accordingly (see whitepaper §10).

11. Authors & references

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The full whitepaper is at [docs/whitepaper/ReineiraOS-Whitepaper.pdf](#); references and formal specifications appear in Appendices A and C, respectively.

Selected references (numbering follows Appendix A of the whitepaper):

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